SELF-ASSEMBLED MELAMINE MICROLENS ARRAYS FOR IMMUNOFLUORESCENCE ENHANCEMENT

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ABSTRACT

Self-assembled dielectric microlenses are used for focusing of the fluorescent signal emitted from a surface-based immunoassay, performed on silane micropatterns as assay substrates, to enhance the detection signal. In our model system, a fluorescent immunocomplex is formed on (3-aminopropyl)triethoxysilane (APTES) microstructures, and then carboxyl-functionalized melamine microparticles with two different sizes are electrostatically self-assembled on the microstructures. Mouse IgG diluted in phosphate buffered saline (PBS) is used as model target antigen and easily detected down to a concentration of 2 ng/mL. We also present a detailed two-dimensional numerical study using the Finite Element Method (FEM).

KEYWORDS: Dielectric microparticles, Immunoassay, Optical detection, Photonic nanojet

INTRODUCTION

Among a variety of bioanalytical immunoassay principles, on-chip 'sandwich' immunoassays, in which the target antigen (Ag) of interest reacts with a capture antibody (Ab) that is grafted on the substrate and subsequently detected via a second fluorescent detection Ab, provides high sensitivity and ease of operation. However, when the fluorescence is quantified with an optical microscopy system, the fluorescent Ab in the microscope focal plane are normally detected on a two-dimensional (2D) flat substrate [1]. Dielectric microparticles with low optical absorbance and high refractive index have been used as *in situ* lenses for focusing the fluorescent signal and improved photon detection [2]. The commercial availability of optically transparent microspheres with many surface functionalizations makes these attractive candidates for optical signal amplification, and they can be used as *in situ* lenses for concentration of the optical signal coming from the substrate [3,4].

EXPERIMENTAL

3 µm-dot and 8 µm-square APTES micropatterns are fabricated on a glass substrate. A glass wafer is spin-coated with positive photoresist, dot and square patterns are generated by a standard photolithography process. Subsequently, 10% (v/v) APTES solution in de-ionized water is spin-coated onto the substrate and then baked for condensation. Afterwards, the photoresist is ultrasonically removed in acetone, resulting in APTES micropatterns on the glass chip, which is then bonded to a polydimethylsiloxane (PDMS) half-channel (120 nL in volume) by mechanically clamped the stack between two poly(methyl methacrylate) plates with access holes for inlet and outlet connections; the former are connected to syringe pumps for generating positive pressure to inject the reagents and microparticles solution in the microchannel. Streptavidin, biotinylated goat anti-mouse IgG (capture Ab), mouse IgG (target Ag), and Alexa Fluor 647-labeled goat anti-mouse IgG (detection Ab) are subsequently flown into the channel, followed by introduction of melamine microparticles (Fig. 1).

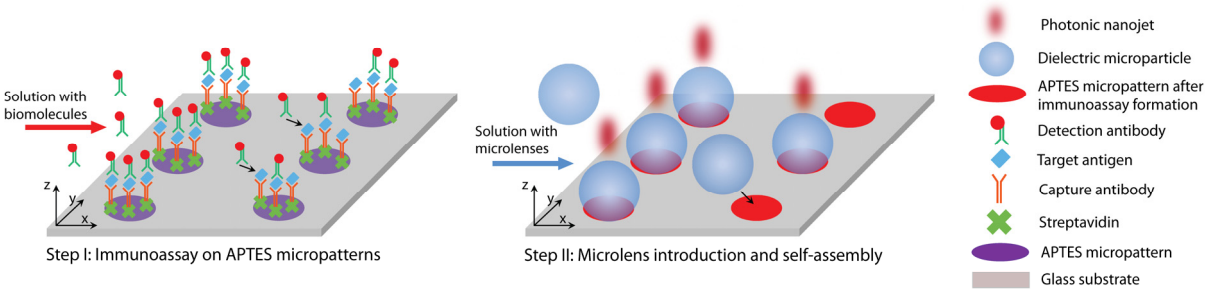

Figure 1: Schematic of a sandwich immunoassay performed in a microfluidic channel on APTES micropatterns (step I), followed by microlens self-assembly (step II).

As the first step, 100 µg/mL streptavidin in PBS is flown into the microfluidic channel at a flow rate of 1 nL/s for 2 min and incubated for 5 min, providing the linker molecules between APTES and later the capture Ab. After briefly washing with 1% (v/v) PBS-Tween buffer, the microchannel is blocked by 1% (w/v) PBS-BSA for 2 min to avoid unspecific adsorption of Abs/Ags on the channel wall. Then, 10 µg/mL capture Ab is injected into the channel and incubated for 5 min. Afterwards, 360 nL of the target Ag diluted in PBS with different concentrations is introduced into the channel by 3 consecutive fill/incubation steps, each taking 2 min for filling and 10 min for incubation. After washing with PBST, 10 µg/mL detection Ab is pumped into the channel and incubated for 5 min. The bright-field and fluorescent images, focusing on 3 µm-dot and 8 µm-square APTES micropatterns, are shown in Fig. 2 (a, b) and Fig. 3 (a, b), respectively. As a final step, 3 µm and 9.75 µm carboxyl-functionalized microparticles based on melamine resin are loaded into

the microfluidic channel. By precisely controlling the flow rate and the incubation time, the electrostatic assembly of microparticles is well-achieved and with good alignment with respect to the APTES microstructures; the bright-field images taken after the microparticle assembly are shown in Fig. 2 (c) and Fig. 3 (c). This technique shows a direct signal enhancement at the focal plane of the microparticles, compared to the signal at the center of the APTES microstructures (Figs. 2 (d) and Fig. 3(d)).

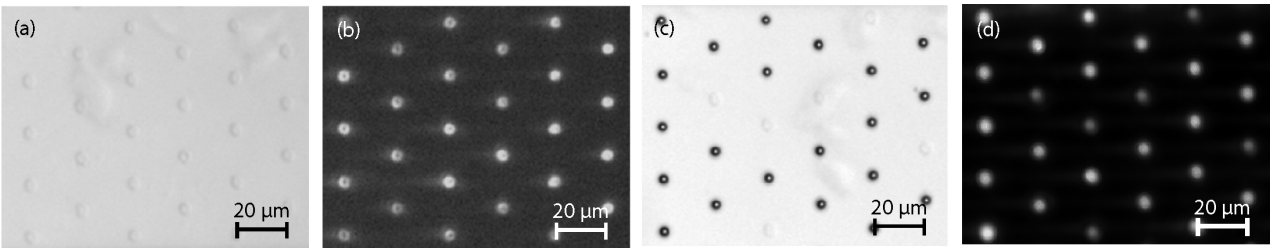

Figure 2: Bright field images (a) before and (c) after the self-assembly of 3 µm microlenses. Fluorescent images are obtained by focusing on (b) the APTES microstructures before the microlens assembly, and on (d) the focal plane of the microlenses.

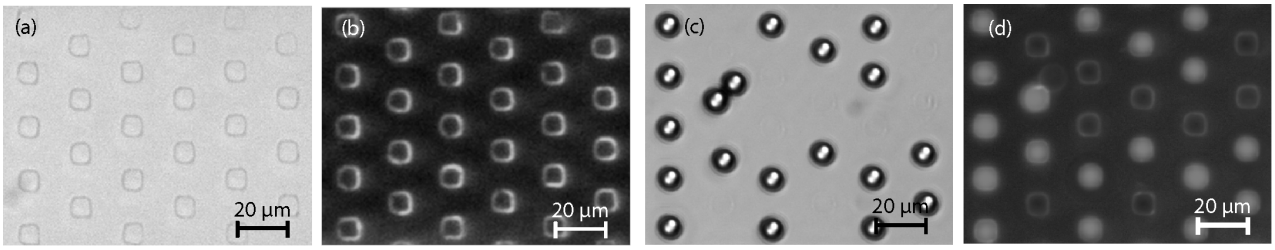

Figure 3: Bright field images (a) before and (c) after the self-assembly of 9.75 µm microlenses. Fluorescent images are obtained by focusing on (b) APTES structures and (d) the microlenses' focal plane.

RESULTS AND DISCUSSION

By comparing the fluorescence sampled at the focal plane of a microlens with the signal originating from the center of an APTES microstructure in absence of the microlens (Fig. 4(a)), we find that application of a 3 µm microlens allows amplifying the immunofluorescence by a factor ~ 2.8. On the other hand, the 9.75 µm microlens shows an amplification factor of ~ 2 (Fig. 4(b)). However, when compared with the signal on the ridge of the square APTES microstructure, the fluorescence intensity on the focal plane of the 9.75 µm microlens is smaller. This implies that the 9.75 µm melamine microparticle cannot be used as an efficient lens to enhance the fluorescent signal. We believe this is because of two reasons: first, when the objective focal plane coincides with the focal plane of the 9.75 µm microlens, it is actually not well focused on the fluorophores, hence, the excitation of the fluorescence is less efficient; second, as the light absorbance in the dielectric medium (the melamine) is proportional to its thickness, the bigger microparticle absorbs more light and there is less signal enhancement. Experiments at different target Ag concentrations are performed. Each point and error bar in Figure 4 correspond to the statistical average and variance from three nominally identical immunoassay experiments. The 0 ng/mL concentration means that the experiment is performed without applying target Ag in PBS. In this case, the signal corresponds to the unspecific adsorption of detection Ab on APTES structures.

Moreover, the numerical study of the electromagnetic field distribution in the vicinity of a dielectric microsphere using FEM is performed and provides insight into the signal amplification mechanism of a microlens. In particular, the software COMSOL Multiphysics is used to study the electromagnetic wave propagation through water, melamine particle with different sizes and glass substrate. A 2D model is built to mimic the experimental situation, in which a microparticle is immobilized on APTES microstructure (on a glass substrate) and immersed in a water- based medium. The APTES structure is simplified as a line source with three components, the central part with weak signal and two edges with strong signal, corresponding to the experimental measurements along the APTES microstructures. A scalar equation is used to study transverse electric wave propagation and light absorbance in melamine particle is also taken into account in the model,

$$\nabla \times \left(\mu_{\mathrm{r}}^{-1} \nabla \times \boldsymbol{E}\right) - \left(\varepsilon_{\mathrm{r}} - \mathrm{j}\sigma / \omega\varepsilon_0\right) k_0^2 \boldsymbol{E} = 0 \qquad (1)$$

$$\varepsilon_{\mathrm{r}} = \left(\mathrm{n} - \mathrm{i}\kappa\right)^2 \qquad (2)$$

As shown in the insert of Fig. 5, the microparticle acts as a microlens focusing the radiation from the substrate into a extremely small region, generating a 'photonic nanojet'. The signal in this region is enhanced: the peak intensity can be several orders of magnitude larger than the intensity of the source. By properly choosing the particle size and its optical

properties, the experimental configuration can be optimized to obtain the best signal amplification factor. As shown in Fig. 5, the maximum intensity is obtained with a 3 µm melamine microparticle.

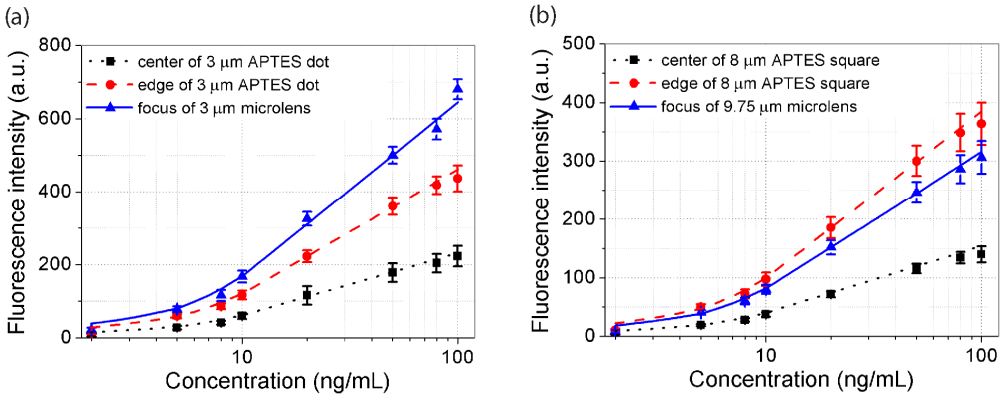

Figure 4: Fluorescence intensity for different target antigen concentrations in combination with (a) 3 µm and (b) 9.75 µm self-assembled microlenses, respectively. The data are obtained by focusing the microscope objective at the focal plane of the microlenses, and at the centre and the edge of the APTES microstructures (in absence of the microlenses). Points represent the average, error bars the variance, while lines are guides to the eye.

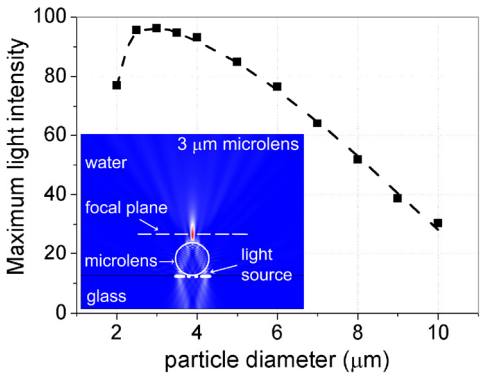

Figure 5: FEM simulation of the maximum intensity on the focal plane of a microlens as a function of the particle size. The dots are values obtained from the simulation, and the dashed line is a guide to the eye. The insert shows the simulation result for a 3 µm microlens.

CONCLUSION

The presented technique shows the enhancement of the immunofluorescence signal by the self-assembled microlenses, as was also verified by the FEM simulations. In future work, microparticles with better optical transparency will be used to generate higher signal collection efficiency. Our approach offers high potential for lowering fluorescent detection limits of surface-based immunoassays.

ACKNOWLEDGEMENTS

The authors would like to thank A.T. Ciftlik and Y. Zhang for discussions. Funding of this work is provided by the Swiss National Science Foundation (Grant 200020-140328).

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